

Abu Hazafa

Glasgow, United Kingdom | (+44) 7831 370 470 | abuhuzaifavirk@gmail.com | ahazafa@unisa.it

UK dependent visa holder; eligible to work in the United Kingdom without restriction | [LinkedIn](#) | [orcid.org](#)

PROFESSIONAL SUMMARY

Computational and translational oncology researcher with a PhD from the University of Salerno and experience spanning multi-omics biomarker discovery, pharmacogenomics, molecular biology, and cancer drug-response research. Skilled in analysing patient-derived samples, cancer models, and clinical laboratory data, with strong scientific communication, organisation, and research documentation abilities. Recognised among the top 2% of researchers worldwide and seeking to apply oncology expertise to clinical research and trial operations.

EDUCATION

University of Salerno

PhD, Translational and Clinical Medicine

Dec 2025

Fisciano, Italy

University of Agriculture Faisalabad

M.Phil, Biochemistry

Dec 2020

Faisalabad, Pakistan

University of Agriculture Faisalabad

M.Sc, Biochemistry

Aug 2018

Faisalabad, Pakistan

University of the Punjab

B.Sc, Zoology, Botany, Chemistry

Aug 2016

Lahore, Pakistan

EXPERIENCE

Ospedali Riuniti San Giovanni di Dio e Ruggi d'Aragona

Researcher in Molecular Pharmacology

Jan 2024 - Oct 2025

Salerno, Italy

- Investigated the UH2/U plasma ratio as a predictive biomarker for fluoropyrimidine toxicity risk in oncology patients.
- Characterised associations between the UH2/U ratio and DPYD gene polymorphisms encoding dihydropyrimidine dehydrogenase to support pharmacogenomic assessment.
- Performed RT-qPCR for SNP identification; conducted DNA/RNA isolation and PBMC extraction from patient samples.

Department of Pharmacy (DIFARMA), University of Salerno

Researcher in Cell Culture and Molecular Biology

Jul 2023 - Dec 2023

Fisciano, Italy

- Evaluated responses to 5-azacytidine, Vemurafenib, Metformin, and Olaparib, individually and in combination, in HNSCC cell lines (SCC-154, CAL-27) targeting CHAF1a/CHAF1b in an oncology research project.
- Applied western blotting, MTT assay, cell migration assay, immunofluorescence, flow cytometry, and transfection; analysed data in GraphPad Prism and R.

Department of Medicine, Surgery and Dentistry (DIPMED), University of Salerno

Doctoral Researcher

Nov 2022 - Nov 2025

Fisciano, Italy

- Designed and executed four simultaneous computational and translational oncology research projects, coordinating activities across multiple study streams.
- Constructed 12 LASSO-selected EMT/DDR prognostic signatures (90 total gene entries across 6 TCGA cancer types) using multi-omics data; performed survival analysis, LASSO-Cox regression, nomogram construction, and immune deconvolution in R, achieving C-index ≥ 0.75 in TCGA-BRCA for both signatures.
- Investigated fluoropyrimidine pharmacogenomics through UH2/U ratio analysis and DPYD polymorphism characterisation in oncology research.
- Profiled drug responses in HNSCC cell lines using a panel of epigenetic and targeted agents.
- Analysed mitochondrial DNA alterations and MitoCarta gene-expression profiles across breast cancer subtypes (luminal, TNBC, HER2+) using machine learning clustering in Python.

Faculty of Science, Cairo University

External Thesis Examiner

2024 - 2025

Cairo, Egypt

- Examined two M.Sc. Biochemistry theses: lncRNA TCONS as a biomarker in idiopathic thrombocytopenic purpura and the anticancer activity of wheatgrass extract on larynx cell lines.
- Evaluated methodology, statistical analysis, and originality; submitted formal assessment reports to the Faculty of Science.

Medsol Clinical Labs

Lab Technologist

Jan 2021 - Dec 2021

Islamabad, Pakistan

- Rotated across biochemistry, hematology, pathology, immunology, and microbiology sections, performing diagnostic laboratory tests.

International Society of Engineering Science and Technology (ISEst)

Research Associate (Remote)

Aug 2018 - Oct 2021

Nottingham, United Kingdom

- Contributed to research and development activities, including writing and reviewing research articles and preparing materials in Microsoft Word and PowerPoint.
- Delivered training on research tools, including EndNote, OriginLab, SPSS, and Microsoft Office applications such as Word, Excel, and PowerPoint.

PROJECTS

EMT/DDR Prognostic Signatures Project

Nov 2022 - Dec 2025

- Developed and externally validated 12 LASSO-selected prognostic signatures across six solid tumors using TCGA training data and independent GEO cohorts, supporting oncology-focused survival analysis.
- Integrated survival modeling, LASSO-Cox regression, nomograms, immune-infiltration profiling, pathway enrichment, mutational analysis, pharmacogenomic prediction, CMap analysis, and WGCNA to evaluate tumor biology and clinical relevance.
- Nomograms improved survival prediction over individual clinical variables based on higher time-dependent ROC AUCs at 1-, 3-, and 5-year overall survival; C-index was optimal (≥ 0.75) in TCGA-BRCA for both signatures and moderate (0.65-0.74) across most other cohorts, supported by calibration, DCA, NRI, IDI, and AIC analyses.
- Analysis scripts: github.com/Hazafa4/EMT_DDR_Project (private; public with DOI on acceptance).

Mitochondrial DNA (mtDNA) Alterations in Breast Cancer

Jan 2025 - Feb 2026

- Built and characterized a mitochondrial gene-based prognostic model in TCGA-BRCA by integrating transcriptomic, clinical, PAM50 subtype, somatic mutation, mtDNA copy-number, and survival data.
- Applied limma-voom, Cox/LASSO-Cox modeling, Kaplan-Meier, ROC, C-index, WGCNA, ssGSEA, TMB/ITH profiling, and mitochondrial genomic feature analysis to assess breast cancer risk and prognosis.
- Identified metabolic risk phenotypes linked to breast cancer prognosis.
- Analysis scripts: github.com/Hazafa4/mtDNA-Project (private; available on request).

Nano-Biopesticides Project

Apr 2020 - Dec 2020

- Evaluated novel pesticidal and antimicrobial activities of natural products formulated in nanosuspension using the anti-solvent precipitation method proposed by Response Surface Methodology (RSM).
- Characterised physicochemical properties of plant extracts and nano-suspensions using Zeta-sizer, FT-IR, and HPLC analysis.

SKILLS

Programming: R (RStudio), Python (NumPy, Pandas, scikit-learn, PyTorch, Biopython, Matplotlib, Seaborn, Jupyter), Bash/Shell scripting, Git/GitHub, R Markdown, MySQL

Bioinformatics Pipelines: RNA-seq analysis (limma, edgeR), Survival analysis (Kaplan-Meier, Cox PH, timeROC), LASSO-Cox regression, Immune profiling (ssGSEA, CIBERSORTX, ESTIMATE, TIDE), Gene-set enrichment (clusterProfiler, GSVA, fgsea, msigdb), Mutation analysis (maftools, MutationalPatterns), WGCNA, PCA, t-SNE

Omics Databases: TCGA, GEO, GTEx, NCBI, cBioPortal, KEGG, STRING, Ensembl, LD-Link, Human Protein Atlas, Roadmap Epigenomics

Visualisation: ggplot2, ComplexHeatmap, patchwork, pheatmap, Cytoscape, Adobe Illustrator, OriginPro, GraphPad Prism

Reproducibility: Zenodo (DOI archival), Jupyter Notebooks, Reproducible workflow documentation

Molecular Techniques: DNA/RNA isolation, RT-qPCR, PCR, SNP genotyping, Western blotting, Protein extraction, proliferation assays, SDS-PAGE, Agarose gel electrophoresis, immunoblotting

Cell Culture and Imaging: 2D/3D mammalian cell culture, PBMC isolation and cryopreservation, MTT/LDH assays, Migration and invasion assays, Transfection (Lipofectamine), Immunofluorescence, Confocal microscopy

Flow Cytometry: Cell surface and intracellular staining, Apoptosis detection

Clinical Research: Oncology/Hematology clinical trials, Clinical trial processes

Microsoft Office Suite: Microsoft Word, Microsoft Excel, Microsoft Project, Microsoft Outlook, Microsoft PowerPoint

AWARDS & HONORS

- **Top 2% of Researchers Worldwide:** Elsevier / Stanford University (2024-2025): Recognised for research impact and citation metrics across all scientific disciplines.
- **Fully-Funded PhD Scholarship (sole competitive position):** University of Salerno (Nov 2022): Three-year award covering tuition, stipend, accommodation, meals, and health insurance.
- **Star of UAF Award for Outstanding Academic Achievement:** University of Agriculture Faisalabad (Apr 2019).
- **Medal of Leadership:** Best Being Services Organization (Jun 2019).

PUBLICATIONS

- Naeem M, Hazafa A, Bano N et al. Explorations of CRISPR/Cas9 for improving the long-term efficacy of universal CAR-T cells in cancer immunotherapy. *Life Sciences*. 2023; 121409. doi:10.1016/j.lfs.2023.121409
- Hazafa A, Batool A, Ahmad S et al. Humanin: a mitochondrial-derived peptide in the treatment of apoptosis-related diseases. *Life Sciences*. 2020; 263:118679. doi:10.1016/j.lfs.2020.118679
- Hazafa A, Mumtaz M, Farooq MF et al. CRISPR/Cas9: a powerful genome editing technique for the treatment of cancer cells. *Life Sciences*. 2020; 263:118525. doi:10.1016/j.lfs.2020.118525
- Hazafa A, Rehman KU, Jahan N, Jabeen Z. The role of polyphenol (flavonoids) compounds in the treatment of cancer cells. *Nutrition and Cancer*. 2020; 72(3):386-397. doi:10.1080/01635581.2019.1637006
- Hazafa A, Rehman KU, Haq IU et al. The broad-spectrum antiviral recommendations for drug discovery against COVID-19. *Drug Metabolism Reviews*. 2020; 52(3):408-424. doi:10.1080/03602532.2020.1770782

- Batool A, Hazafa A, Ahmad S et al. Treatment of lymphomas via regulating signal transduction pathways by natural therapeutic approaches. *Life Sciences*. 2021; 269:119064. doi:10.1016/j.lfs.2021.119064
- Naeem M, Iqbal MO, Khan H, Hazafa A et al. Twenty years of research on the regulation of signaling pathways by natural products in breast cancer. *Molecules*. 2022; 27(11):3412. doi:10.3390/molecules27113412
- Sodagar A, Javed R, Hazafa A et al. Pathological features and neuroinflammatory mechanisms of SARS-CoV-2 in the brain. *Biomolecules*. 2022; 12(7):971. doi:10.3390/biom12070971

MANUSCRIPTS UNDER REVIEW / IN PREPARATION

- *1. Hazafa A, Campana F, Rosa V, Iommelli F, Lonardo E, Rosa C, Corte CMD, Petrillo A, Filippelli A, Conti V, Vecchio VD, Noce ML, Papaccio L. **Cross-Cancer Epithelial–Mesenchymal Transition (EMT) and DNA Damage Response (DDR) Signatures Define High-Risk Phenotypes Associated with Immune Evasion**. Target: Cell Reports Medicine. In preparation, 2026.
- *2. Hazafa A et al. Mitochondrial DNA alterations and MitoCarta gene expression profiles across breast cancer molecular subtypes. In preparation, 2026.

SCIENTIFIC WRITING AND PEER REVIEW

- * Reviewed >200 manuscripts for indexed journals in oncology, cell biology, biochemistry, molecular biology, and pharmacology.
- * Early Career Editorial Board Member: Experimental Results (Cambridge University Press); Oncology Research (IF: 4.1/Q1) (Tech Science Press).
- * Proficient in Zotero and EndNote for reference management and systematic review workflows.

PROFESSIONAL MEMBERSHIPS

<i>Sep 2025 – present</i>	<i>Biochemical Society (Postgraduate Membership)</i>	<i>ID: 30006866</i>
<i>Nov 2025 – present</i>	<i>European Association for Cancer Research (EACR)</i>	<i>ID: EACR42732</i>
<i>Oct 2025 – present</i>	<i>American Association for Cancer Research (AACR)</i>	<i>ID: 2626147</i>
<i>Jan 2025 – present</i>	<i>Society for Immunotherapy of Cancer (SITC)</i>	<i>ID: 153855</i>
<i>Feb 2024 – present</i>	<i>American Society of Clinical Oncology (ASCO)</i>	<i>ID: 7383052</i>
<i>Nov 2023 – Dec 2025</i>	<i>American Physiological Association (APA)</i>	<i>ID: C2103081734</i>

SELECTED CONFERENCES AND SEMINARS

- * ISSCR Annual Meeting 2024, International Society for Stem Cell Research. Hamburg, Germany, July 2024.
- * Evolution of Cancer Pharmacology, Italian Society of Pharmacology. Salerno, March 2024.
- * Finding the Cure for Cancer Through Research Series (AIRC), 2024-2025: Pancreatic cancer heterogeneity (Jan 2024); Metabolic reprogramming in renal carcinoma (Mar 2024); Haematological malignancies and immunity (Jun 2024); Immunometabolic dysfunctions in cancer (May 2025).
- * Translating Discoveries into Clinical Practice in Colorectal Cancer, AIRC. October 2023.
- * Challenges in 3D Cell Culture and Phenotypic Analysis, Bitesize Bio. July 2023.
- * The Diary of a Scientist Webinar Series, BioMed X. June 2025.

REFERENCES

Prof. Rais A. Ansari - Dept of Pharmaceutical Sciences, Nova Southeastern University, Fort Lauderdale, FL 33328, USA.
ra557@nova.edu | +1 (954) 859-8483
 Dr Federica Papaccio - Dept of Medicine, Surgery, and Dentistry, University of Salerno, Fisciano 84084, Italy.
fpapaccio@unisa.it

CERTIFICATION

- Certificate in Cancer Immunotherapy: (8 modules, 17 CPD hours) - Society for Immunotherapy of Cancer (SITC), Oct 2025 (Virtual)
- SITC Cancer Immunotherapy Winter School: Society for Immunotherapy of Cancer (SITC), Feb 2025 (Virtual): Tumour immunology; biomarker validation; clinical trial design; combination therapies
- Human Intestinal Organoid Live Virtual Training: (6.6 CPD points) - STEMCELL Technologies, Jun 2025 (Virtual)
- Unconscious Bias in Medicine: Stanford Medicine, Feb 2024 (Online)
- Spring School on Transferable Skills: (2 ECTS) - University of Naples Federico II, May 2024 (Naples, Italy)
- The Science of Medicines: Monash University, Jun-Jul 2020 (Online)